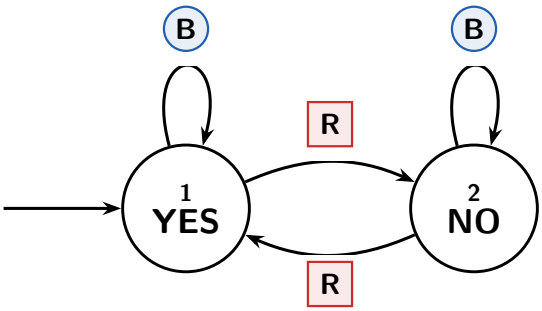
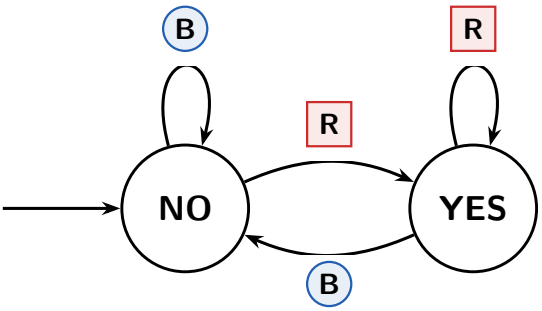
R means red; B means blue. A machine has one start circle and a fixed YES or NO on each circle. Each circle has exactly one R arrow and one B arrow. Read left to right with one marker, hiding used cards. Only the current circle and the card being read may affect the next move.

Problem 1: Find every four-card row that makes exactly one of these machines say YES. Start both machines fresh for each row, and write each row once.





Left machine, input B R: start 1 $\xrightarrow{B}$ 1 $\xrightarrow{R}$ 2; stop: NO.

Problem 2: Draw a machine that says YES exactly when at least one red card has appeared. Use as few circles as you can.

Problem 3: Draw a machine that says YES exactly when its last card is blue. It must say NO if there are no cards. Use as few circles as you can.

Problem 4: A row gets YES exactly when its red cards can be put in groups of three with none left over, including no red cards. For each pair, add the same extra cards in the same order to both rows to give one YES and one NO, or explain why this cannot be done. Adding no cards is allowed.

no cards

R

\+ _____

YES / NO

YES / NO

R

R R

\+ _____

YES / NO

YES / NO

R R

R R R

\+ _____

YES / NO

YES / NO

R B R

R R

\+ _____

YES / NO

YES / NO

B

R R R

\+ _____

YES / NO

YES / NO

R B

B R

\+ _____

YES / NO

YES / NO
